# Generalized Linear Model: Assignment 2

Student ID: Name:

1. Labeling Index (LI) is an indicator of cellular activity measured after the administration of tritiated thymidine to a patient. The following table shows, for different LI values, the number of patients who underwent cancer surgery (Cases) and the number of those who recovered (Remissions; 1 = Yes). Using the data below, perform a logistic regression analysis to examine the relationship between recovery after cancer surgery and LI.

| LI | No. of Cases | No. of Remissions | LI | No. of Cases | No. of Remissions | LI | No. of Cases | No. of Remissions |
|---|---|---|---|---|---|---|---|---|
| 8 | 2 | 0 | 18 | 1 | 1 | 28 | 1 | 1 |
| 10 | 2 | 0 | 20 | 3 | 2 | 32 | 1 | 0 |
| 12 | 3 | 0 | 22 | 2 | 1 | 34 | 1 | 1 |
| 14 | 3 | 0 | 24 | 1 | 0 | 38 | 3 | 2 |
| 16 | 3 | 0 | 26 | 1 | 1 | | | |

a. Calculate the estimated probability of recovery ($\pi$) and the 95% confidence interval for patients with LI = 8 and LI = 26.

b. Calculate the rate of change of $\pi$ for LI = 8 and LI = 26.

c. Compute the 95% confidence interval of the odds ratio for the effect of LI.

d. To perform a likelihood ratio test for the effect of LI, obtain the likelihood under the null hypothesis and the likelihood under the alternative hypothesis, and then conduct the likelihood ratio test.

2. Hastie and Tibshirani 1990, p. 282 described a study to determine risk factors for kyphosis, severe forward flexion of the spine following corrective spinal surgery. The age in months at the time of the operation for the 18 subjects for whom kyphosis was present were 12,15, 42, 52, 59, 73, 82, 91, 96, 105, 114, 120, 121, 128, 130, 139, 139, 157 and for 22 of the subjects for whom kyphosis was absent were 1, 1, 2, 8, 11, 18, 22, 31, 37, 61, 72, 81, 97, 112, 118, 127, 131, 140, 151, 159, 177, 206.

a. Fit a logistic regression model using age as a predictor of whether kyphosis is present. Test whether age has a significant effect.

b. Plot the data. Note the difference in dispersion on age at the two levels of kyphosis.

c. Fit the model $logit[\pi(x)] = \alpha + \beta_1 x + \beta_1 x^2$. Test the significance of the squared age term, plot the fit, and interpret.

3. Let $Y$ denote a subject's opinion about current laws legalizing abortion (1=support), for gender $h$ ($h = 1$, female; $h = 2$, male), religious affiliation $i$ ($i = 1$, Protestant; $i = 2$, Catholic, $i = 3$, Jewish), and political party affiliation $j$ ($j = 1$, Democrat; $j = 2$, Republican; $j = 3$, Independent). For survey data, software for fitting the model

$$logit[P(Y = 1)] = \alpha + \beta_h^G + \beta_i^R + \beta_j^P$$

reports $\hat{\alpha} = 0.62, \hat{\beta}_1^G = 0.08, \hat{\beta}_2^G = -0.08,\ \hat{\beta}_1^R = -0.16, \hat{\beta}_2^R = -0.25, \hat{\beta}_3^R = 0.41, \hat{\beta}_1^P = 0.87, \hat{\beta}_2^P = -1.27, \hat{\beta}_3^P = 0.40$.

a. Interpret how the odds of support depends on religion.

b. Estimate the probability of support for the group most (least) likely to support current laws

c. If, instead, parameters used constraints $\beta_1^G = \beta_1^R = \beta_1^P = 0$, report the estimates.

4. A study for several professional sports of the effect of a player's draft position $d$ $(d = 1,2,3, \ldots)$ of selection from the pool of potential players in a given year on the probability $\pi$ of eventually being named an all-star used the model $logit(\pi) = \alpha + \beta log d$ (S.M. Berry, *Chance*, **14**:53-57,2001).

a. Show that $\pi/(1-\pi) = e^{\alpha} d^{\beta}$.

b. In the United States, Berry reported $\hat{\alpha} = 2.3$ and $\hat{\beta} = -1.1$ for pro basketball and $\hat{\alpha} = 0.7$ and $\hat{\beta} = -0.6$ for pro baseball. This suggests that in basketball a first draft picks with high $d$ are relatively less likely to be all-stars. Explain why.

5. Following Table, refers to the effectiveness of immediately injected or $1\frac{1}{2}$ -hour-delayed penicillin in protecting rabbits against lethal injection with $\beta$-hemolytic streptococci.

| Penicillin Level | Delay | Response | |
|---|---|---|---|
| | | Cured | Died |
| $\frac{1}{8}$ | None | 0 | 6 |
| | $1\frac{1}{2}$h | 0 | 5 |
| $\frac{1}{4}$ | None | 3 | 3 |
| | $1\frac{1}{2}$h | 0 | 6 |
| $\frac{1}{2}$ | None | 6 | 0 |
| | $1\frac{1}{2}$h | 2 | 4 |
| 1 | None | 5 | 1 |
| | $1\frac{1}{2}$h | 6 | 0 |
| 4 | None | 2 | 0 |
| | $1\frac{1}{2}$h | 5 | 0 |

a. Let $X =$ delay, $Y =$ whether cured, and $Z =$ penicillin level. Fit the logit model

$$logit(\pi_{ik}) = \alpha + \beta x_i + \beta_k^Z (i = 1,2, k = 1, \dots ,5)$$

Argue that the pattern of 0 cell counts suggests that (with no intercepts) $\hat{\beta}_1^Z = -\infty$ and $\hat{\beta}_5^Z = -\infty$. What does your software report?

b. Using the logit model, conduct the likelihood-ratio test of $XY$ conditional independence and interpret.

c. Test $XY$ conditional independence using the Cochran Mantel Haenszel test and interpret.

d. Estimate the $XY$ conditional odds ratio using (i) ML with the logit model, and (ii) the Mantel Haenszel estimate and interpret.

e. The small cell counts make large-sample analyses questionable. Conduct small sample inference and interpret.

6. Consider following Table, from a study of nonmetastatic osteosarcoma (A. M. Goorin, *J. Clin Oncol*. **5**: 1178 1184, 1987, and the manual for *LogXact*). The response is whether the subject achieved a three-year disease-free interval.

| Lymphocytic Infiltration | Gender | Osteobliastic Pathology | Disease-Free | |
|---|---|---|---|---|
| | | | Yes | No |
| High | Female | No | 3 | 0 |
| | | Yes | 2 | 0 |
| | Male | No | 4 | 0 |
| | | Yes | 1 | 0 |
| Low | Female | No | 5 | 0 |
| | | Yes | 3 | 2 |
| | Male | No | 5 | 4 |
| | | Yes | 6 | 11 |

a. Show that each predictor has a significant effect when used individually without the others.

b. Try to fit a main-effects logistic regression model containing all three predictors. Explain why the ML estimate for the effect of lymphocytic infiltration is infinite.

c. Using conditional logistic regression, (i) conduct an exact test for the effect of lymphocytic infiltration, controlling for the other variables; and (ii) find a 95% confidence interval for the effect. Interpret results.

7. Suppose that $\{\pi_{ijk}\}$ in a $2 \times 2 \times 2$ table is, by row, (0.15, 0.10 / 0.10, 0.15) when $Z = 1$ and (0.10, 0.15 / 0.15, 0.10) when $Z = 2$. For testing conditional $XY$ independence with logit models having $Y$ as a response, explain why the likelihood-ratio test comparing models $X + Z$ and $Z$ is not consistent but the likelihood-ratio test of fit of the $XY$ conditional independence model is.